Random Error and Confidence Intervals

APHS 2025

Question: What Is Error?

When we talk about error in epidemiology, we are generally talking about the difference between the true underlying population value of interest and our measure (or estimate) of that value.

A. True
B. False

True. Error is simply the difference between the truth and our estimate of the truth.

The Big Picture: What Is Truth?

$$\text{Truth} = \text{Measure} - \text{Error}$$

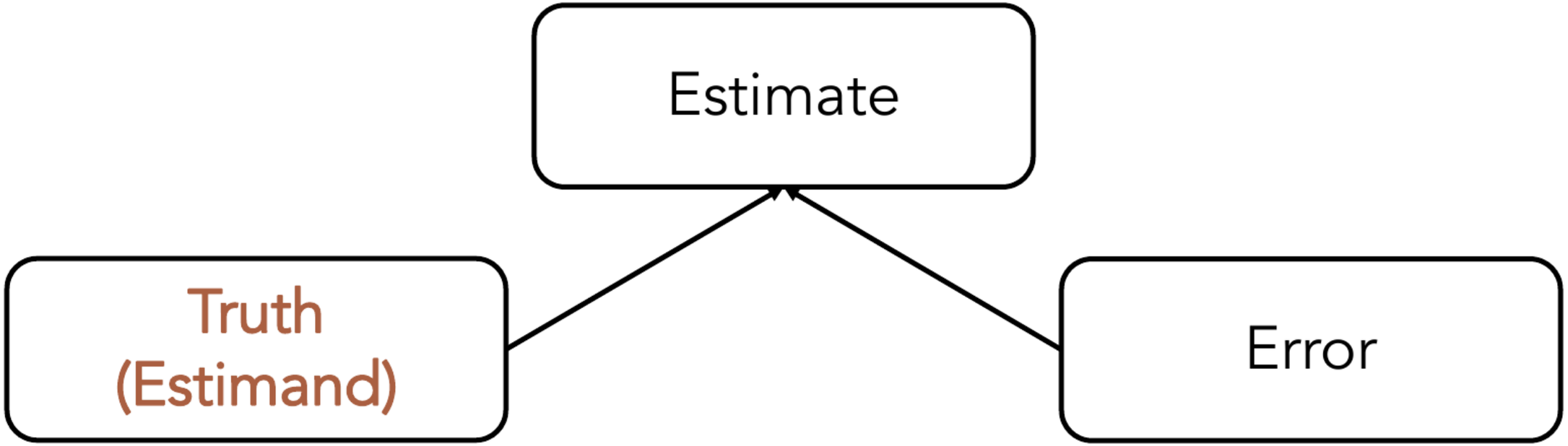
The Big Picture: Truth by Another Name

Truth = Measure - Error

Estimand

Estimate

The Big Picture: Truth from Another Perspective

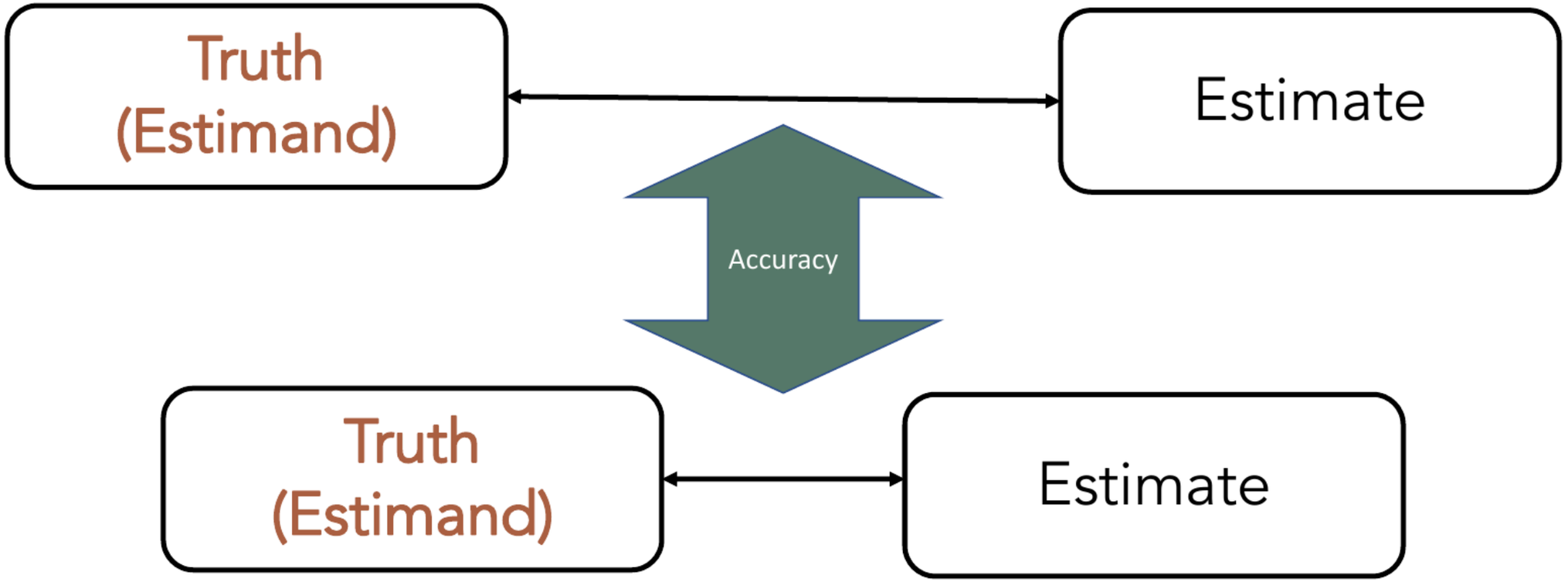


Quick Terminology Review

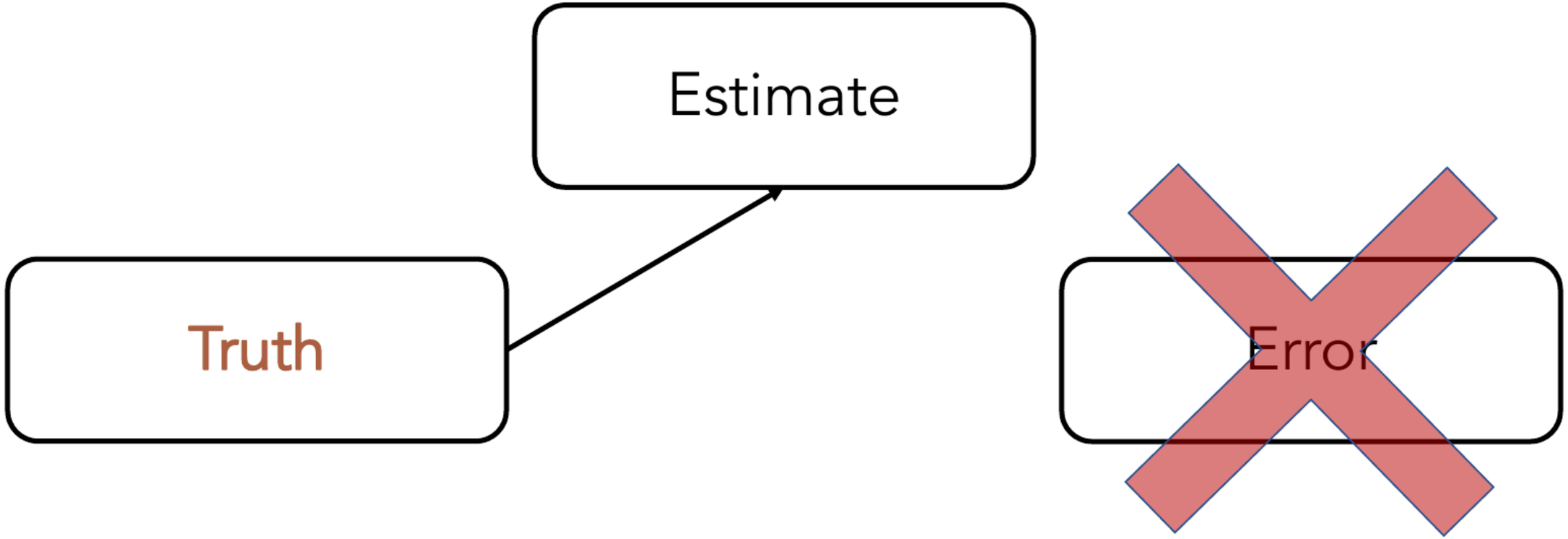
Match each term to its definition.

Terms	Definitions
1. Estimand	A. The quantity that actually resulted from our data collection and analysis.
2. Estimator	B. The difference between the estimate and the estimand.
3. Estimate	C. The true quantity we aspire to measure in our target population.
4. Error	D. The analysis technique or algorithm used to produce the estimate from our data.

The Big Picture: Accuracy



The Big Picture: Estimate without Error



Quick Review: Types of Error

The two highest-level (most general) categories of error are typically considered to be...

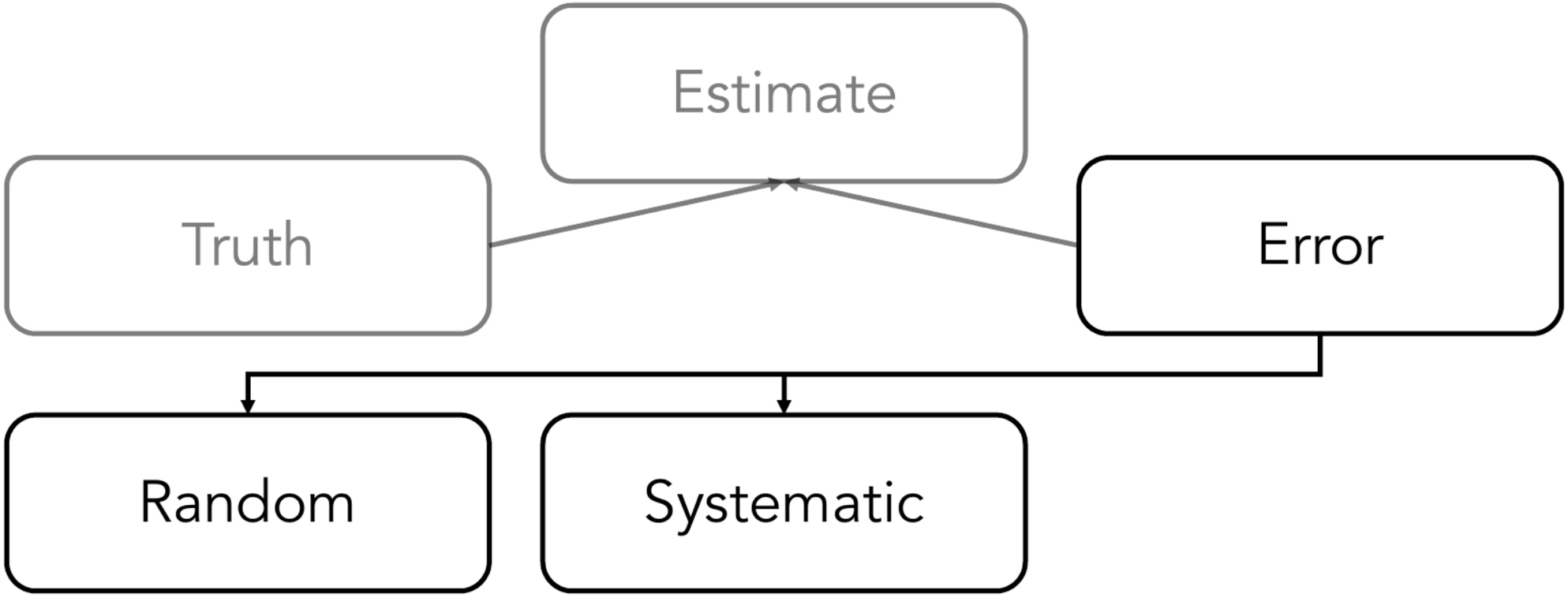
A. Bias and confounding

B. Random and systematic

C. Differential and non-differential

D. Ignorable and non-ignorable

The Big Picture: Types of Error



What is random error?

Thought Experiment: Chance and Predictions

If we just had enough information about causes and effects, then we could perfectly predict every outcome.

A. True

B. False

C. Not sure

How Stressed Are Students?

A campus wellness team wants to estimate the **mean stress score among all undergraduates this semester**.

- **Population:** All undergraduates this semester.
- **Parameter:** Their population mean StressScore.
- **Sample:** 12 randomly selected students.

Should we expect the mean stress score from these 12 students to be the same as the mean for all undergraduates?

Explore the Population in Excel

Open the Excel sampling demonstration and select the **Population** worksheet.

- Each row represents one student in a **simulated population** of 10,000 students.
- These are computer-generated example StressScores, not actual student records.
- Calculate the mean of all 10,000 scores: `=AVERAGE(B7:B10006)`.

Population mean: 20.99 points.

Explore One Sample in Excel

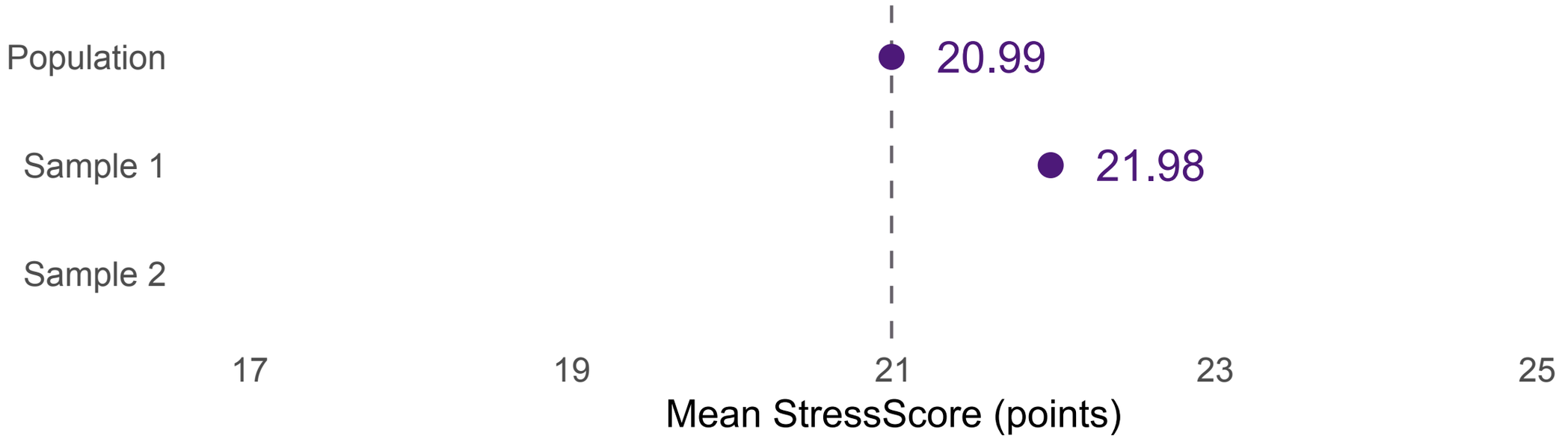
Select the **Samples** worksheet and locate **Sample 1**.

- This random sample contains **12 students** from the simulated population.
- Find their student IDs and StressScores.
- Calculate the mean of their 12 scores.

Sample mean: 21.98 points.

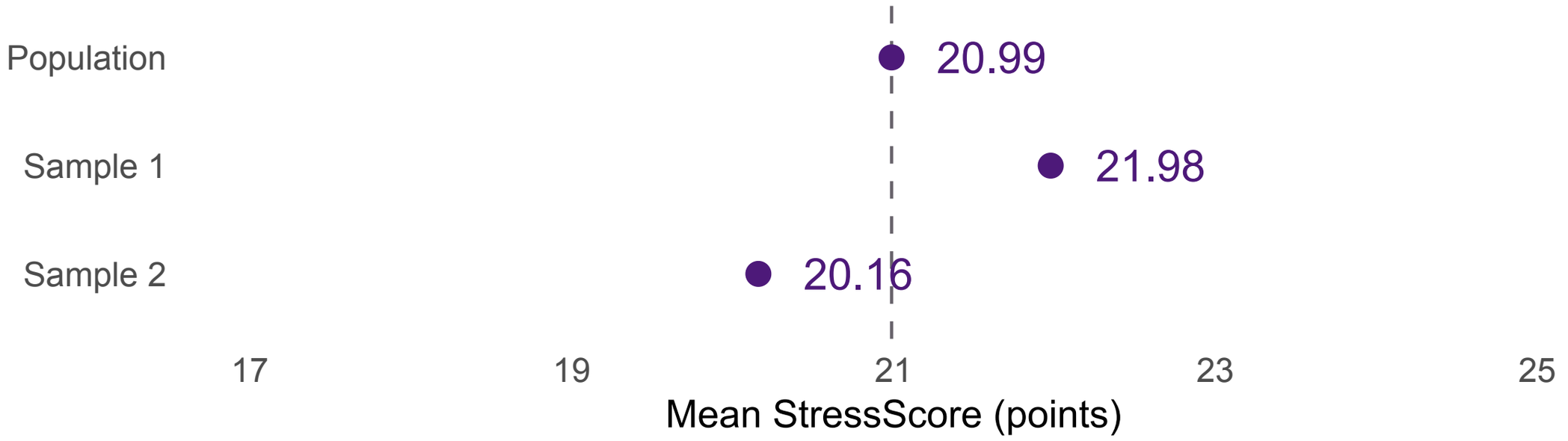
One Population, One Sample

The sample mean is an estimate of the population mean.



The Same Population, Another Sample

Sample 2 contains **another random selection of 12 students** from the same population.



The population stays the same. The sample estimate changes.

In a Real Study, the Population Mean Is Unknown

What We Observe	What We Want to Know
Sample 1's mean StressScore: **21.98 points**	The mean among all undergraduates: **?**

How much might an estimate vary just because we sampled different students?

Sampling Error

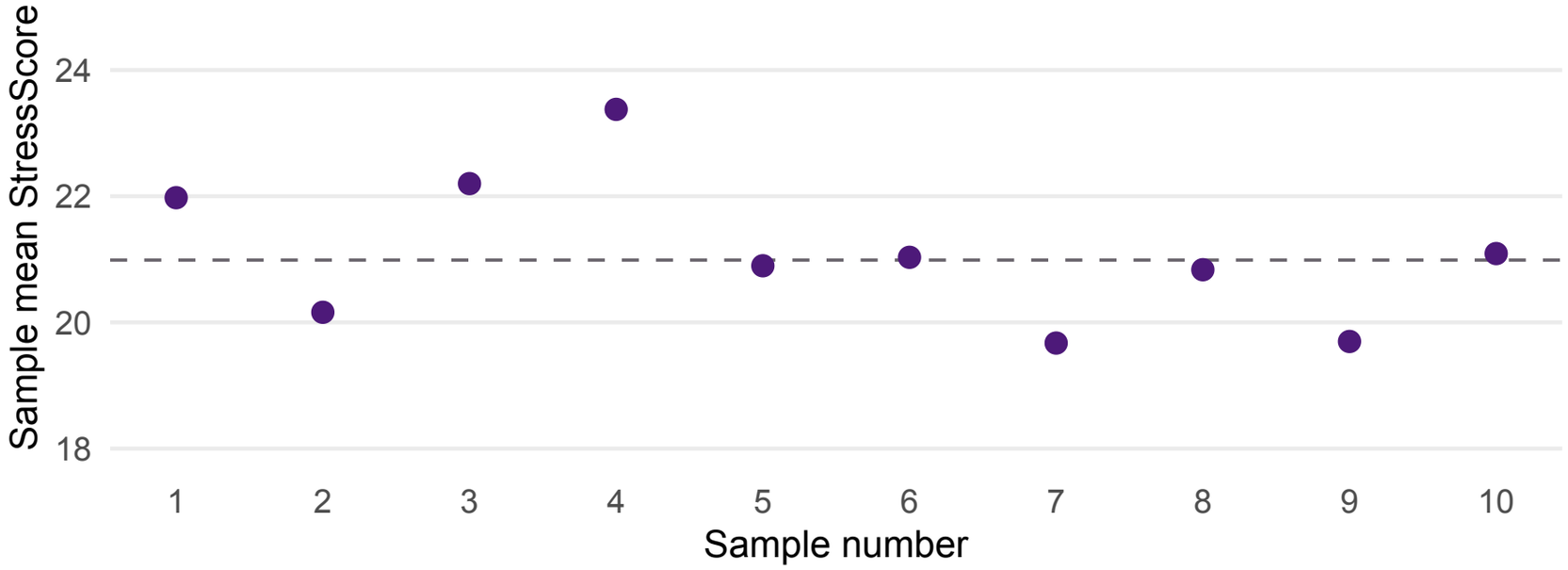
$$\text{Sampling error} = \text{Sample mean} - \text{Population mean}$$

Sample	Sample Mean	Population Mean	Sampling Error
1	21.98	20.99	+0.99
2	20.16	20.99	−0.83

A positive error is an **overestimate**. A negative error is an **underestimate**.

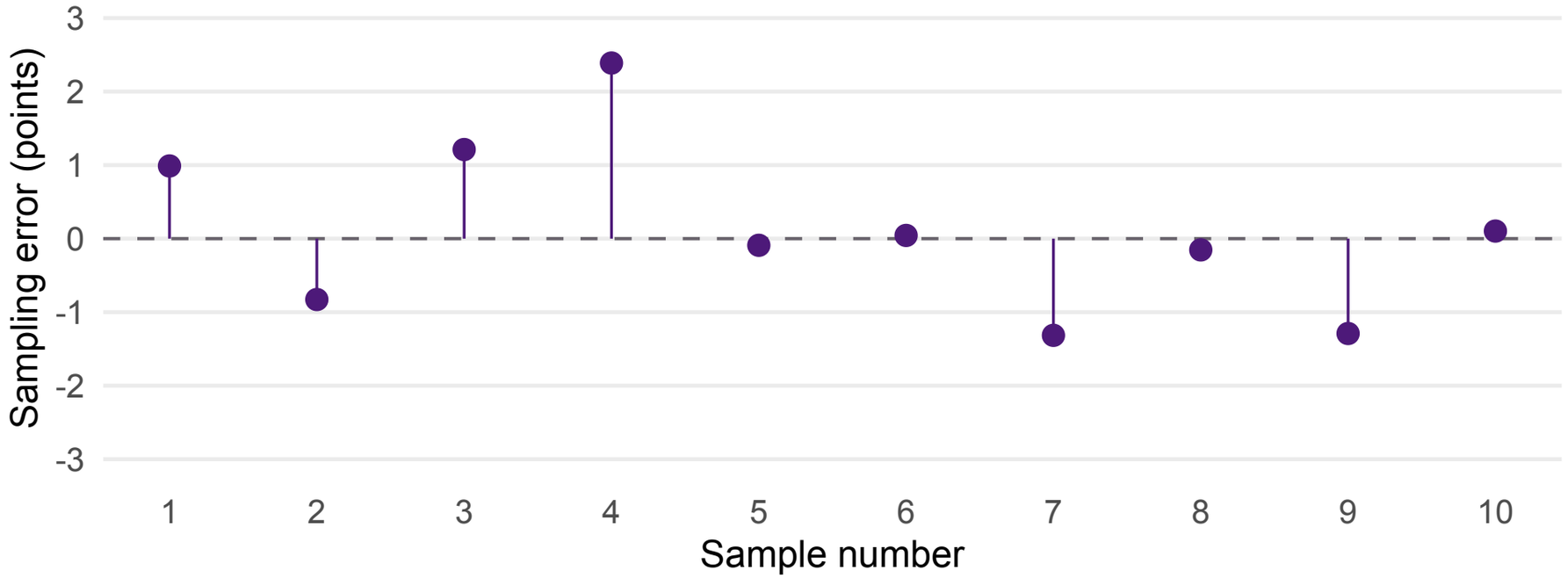
Repeated Samples, Different Means

These are the **first 10 samples** in the Excel sampling demonstration. Each has **12 students**.



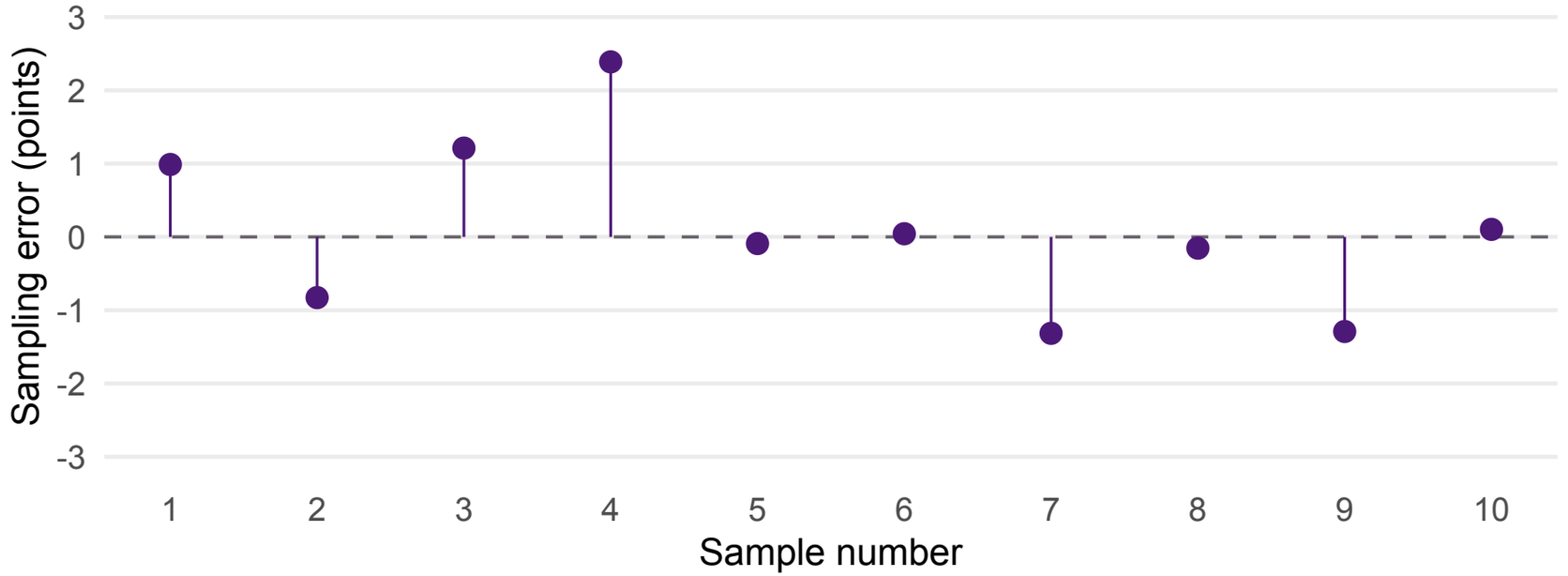
The dashed line marks the **population mean of 20.99**.

The Same Samples, Shown as Errors



Subtract the **population mean** from each sample mean. The dashed line now marks **zero error**.

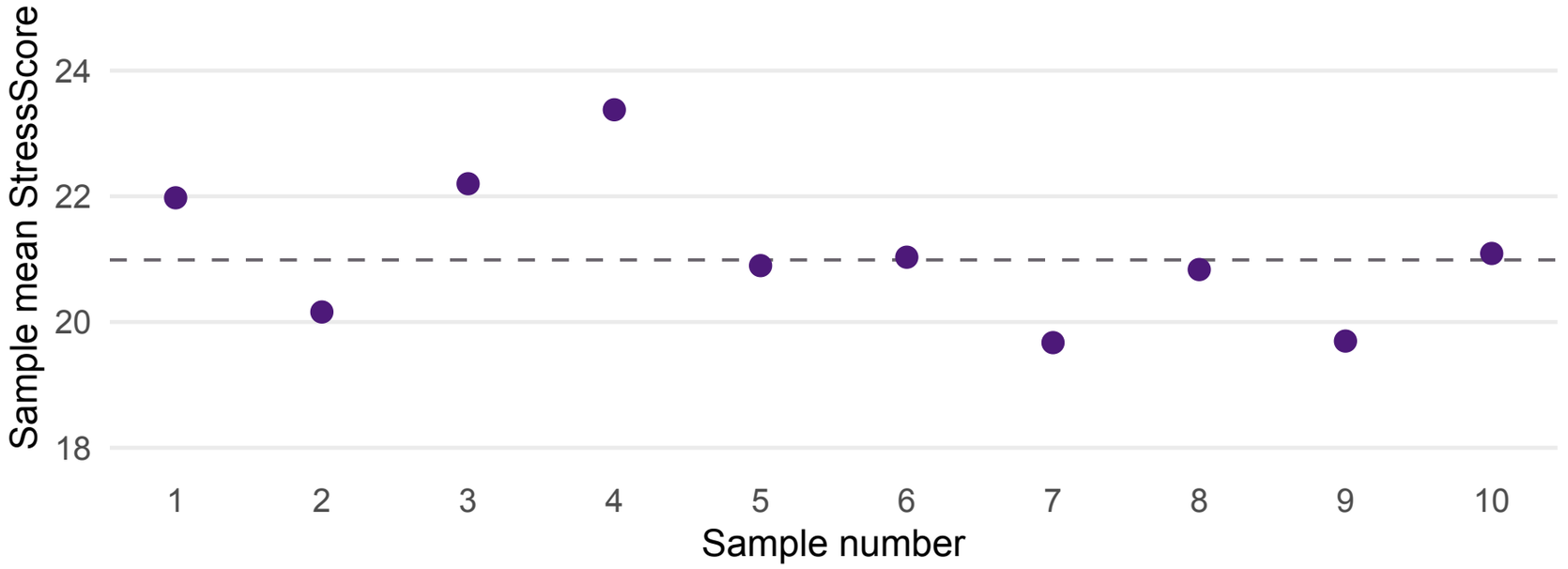
Do the Errors Cancel Out?



For these 10 samples: **sum of errors ≈ +1.06 points**; **average error ≈ +0.106 points**.

The errors in a particular set of samples **need not sum to zero**.

Average of the Sample Means



Average of the 10 sample means ≈ 21.09 points. Population mean ≈ **20.99 points**.

An unbiased estimator has **zero expected error** across repeated samples.

How Much Might Our Estimate Vary?

In our demonstration, we knew the population mean, so we could calculate each sample's error.

In a real study, we usually have **one sample** and an **unknown population mean**.

We cannot calculate our sample's exact error, but we can estimate how much sample means would vary across repeated samples.

- The **standard error** summarizes that variability.
- A **confidence interval** uses it to express uncertainty around our estimate.

Standard Deviation and Standard Error

For the **separate Prep 4 worked sample** of 12 students:

Measure	What Varies?	StressScore Example
Standard deviation	Individual scores within the sample	3.64 points
Standard error	Sample means across repeated samples	1.05 points

For a sample mean, estimate the standard error as **sample SD ÷ $\sqrt{n}$**.

What Is a Confidence Interval?

Our sample mean is **21 points**. It estimates the population mean, but another sample would give us a different estimate.

A **confidence interval** gives a range of plausible values for the population mean, based on our sample and the method's assumptions.

- It expresses uncertainty due to random sampling.
- Its width tells us about the estimate's **precision**: a narrower interval gives a more precise estimate.
- A **95% confidence level** means that, over many repeated samples, about 95% of intervals constructed using this method would contain the true population mean.

What Is the Margin of Error?

For a population mean, we will construct a confidence interval by adding and subtracting a margin of error from the sample mean.

$$\text{Sample mean} \pm \text{Margin of error}$$

The **margin of error** is the distance from the sample mean to either endpoint.

For our worked example:

$$21.00 \pm 2.31 = 18.69 \text{ to } 23.31$$

A larger margin of error produces a wider confidence interval.

Calculating a Confidence Interval for a Population Mean

The margin of error is the **standard error multiplied by a critical value**. The critical value determines how many standard errors we extend on either side of the sample mean.

$$\text{Sample mean} \pm \underbrace{t^* \times \frac{s}{\sqrt{n}}}_{\text{Margin of error}}$$

- $s/\sqrt{n}$ estimates the **standard error**, using the sample SD and sample size.
- Because we estimate the population SD using the sample SD, we use a critical value, t^*, from the **t distribution**.
- The critical value depends on the confidence level and sample size. For **12 students** and **95% confidence**, $t^* \approx 2.20$.

Excel Demo: Repeated Confidence Intervals

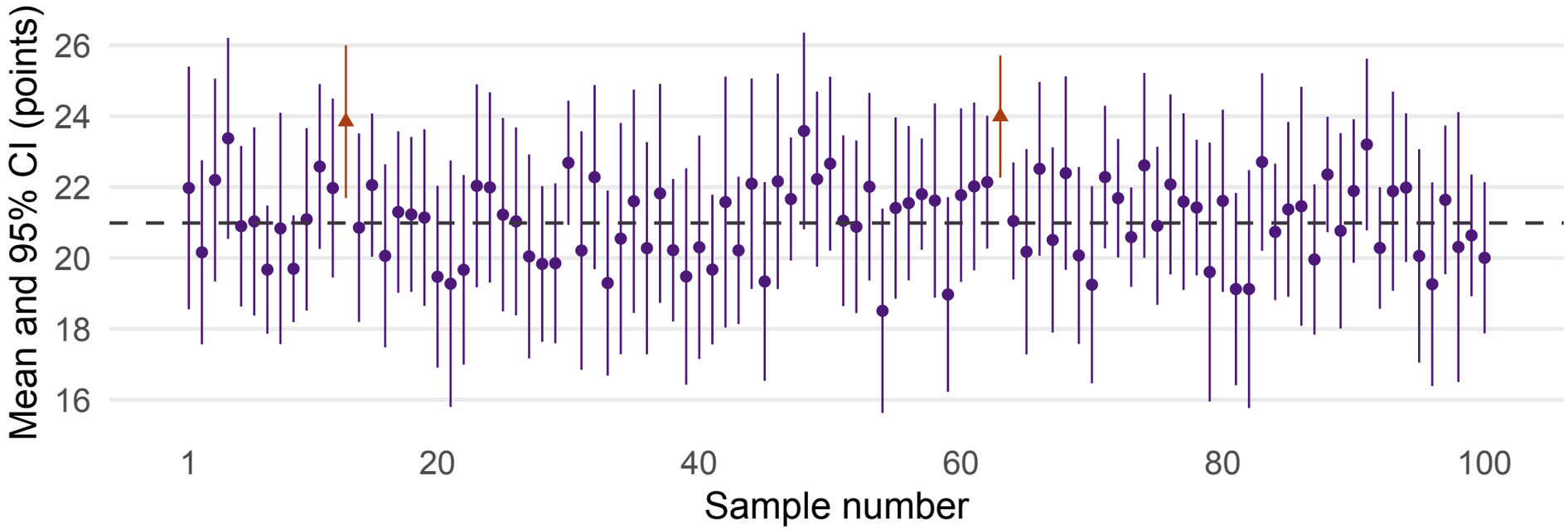
Open the prepared Excel sampling demonstration.

1. On **Samples**, trace one row from its 12 scores to its mean, SE, and interval.
2. Check whether its interval contains the **population mean of 20.99**.
3. On **Coverage**, compare the intervals from all **100 samples**.

Will every interval capture the mean? Must exactly 95 of 100 capture it?

Which Intervals Capture the Population Mean?

Dots show sample means. Vertical lines show **95% confidence intervals**.



98 of 100 capture the mean. Samples **13 and 63** miss it.

Question: What Does 95% Confidence Mean?

A team reports a 95% confidence interval for mean StressScore. What does **95%** describe?

A. The percentage of students whose scores fall inside the interval.
B. The long-run percentage of intervals from this method that capture the true mean.
C. The probability that the true mean is inside this particular interval.
D. The percentage of future sample means that fall inside this interval.

Question: Confidence Level and Interval Width

The team uses the **same 12 scores** and the **same method** to calculate three confidence intervals. Which interval is widest?

A. The 90% confidence interval.
B. The 95% confidence interval.
C. The 99% confidence interval.
D. They all have the same width.

What Makes an Estimate More Precise?

A smaller **standard error** means the sample estimate tends to vary less across repeated samples.

- With the same sample SD and confidence level, a larger sample gives a **narrower** mean confidence interval.
- With the same sample size and confidence level, a larger sample SD gives a **wider** interval.
- For the same data, a higher confidence level gives a **wider** interval.

Prep 4 Worked Example: Mean and Standard Error

Let's calculate a 95% confidence interval from one sample, step by step. Open the Prep 4 worked example.

Calculation	Excel Formula	Result
Sample size	`=COUNT(A5:A1000)`	12
Sample mean	`=AVERAGE(A5:A1000)`	21.00
Sample SD	`=STDEV.S(A5:A1000)`	3.64
Standard error	`=D13/SQRT(D11)`	1.05

Prep 4 Worked Example: Confidence Interval

Calculation	Excel Formula	Result
Degrees of freedom	`=D11-1`	11
Two-sided t value	`=T.INV.2T(0.05,D15)`	2.20
Margin of error	`=D16*D14`	2.31
Lower limit	`=D12-D17`	18.69
Upper limit	`=D12+D17`	23.31

Report the **95% CI for the population mean StressScore as 18.69 to 23.31 points**. Excel uses full precision even when the cells display two decimals.

Prep 4 Worked Example: Comparing the Interval With the Population Norm

Prep 4, Step 3 asks whether your population mean could equal a specified reference value. For practice, the assignment uses 20 for StressScore and 10 for AnxietyScore. These are hypothetical values.

Assigned Outcome	Hypothetical Population Norm
StressScore	20
AnxietyScore	10

The estimated mean is **21 points**, with a **95% CI of 18.69 to 23.31**. The interval includes the **norm of 20**, along with values below and above it.

The data are compatible with a population mean of 20 under the method's assumptions. **This does not establish that the true mean is 20.**

Group Exercise: One Score Changes

In a copy of the **Prep 4 worked workbook**, change the last score from **27 to 31**.

1. Predict what will happen to the mean and to the interval width.
2. Record the new mean, sample SD, and 95% confidence interval.
3. Compare the new interval with the original. Is the estimate more or less precise? Does your decision about the norm of 20 change?

Group Debrief: One Score Changes

Result	Original Scores	With 31 Replacing 27
Sample mean	21.00	21.33
Sample SD	3.64	4.36
95% CI	18.69 to 23.31	18.57 to 24.10

The interval is **wider** because the scores vary more and the sample size is unchanged. The norm of **20** is still inside the interval.

What to Report for Prep 4

- Report the sample size, sample mean, sample SD, and 95% confidence interval for your assigned outcome.
- Explain what the interval estimates and what its width says about precision.
- Compare the interval with the assigned hypothetical norm and explain your decision.